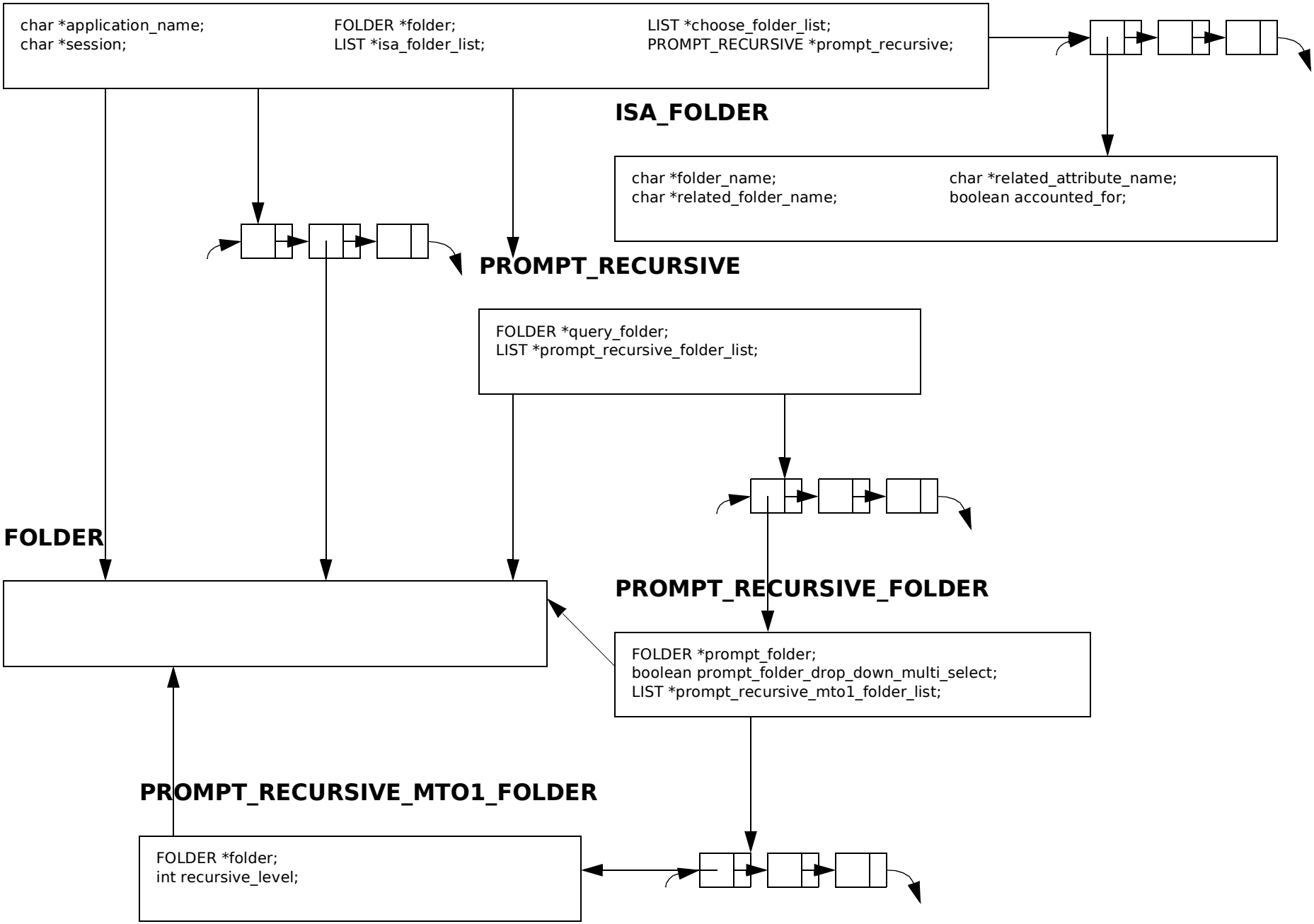
APPASERVER Data Structure

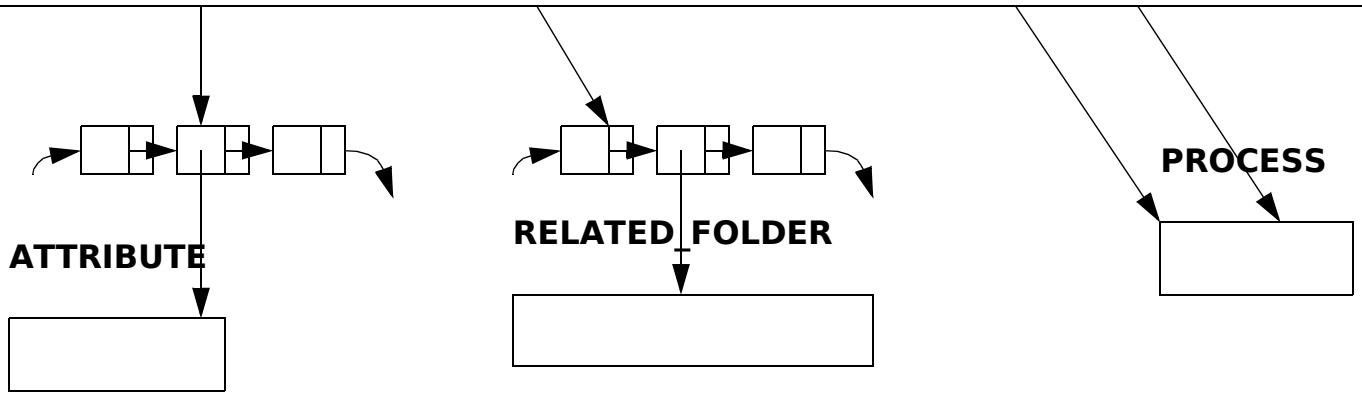
APPASERVER



APPASERVER Data Structure

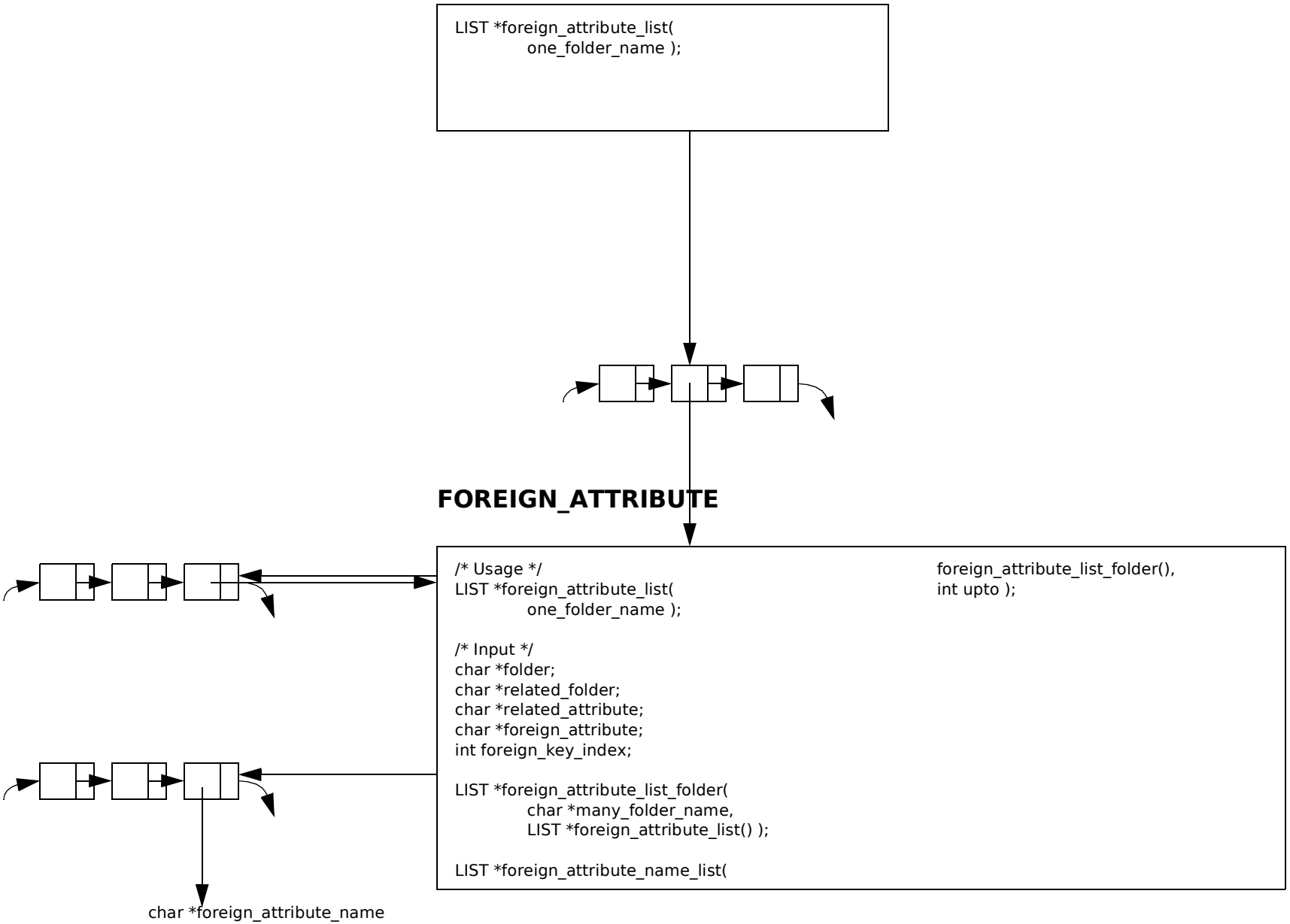
FOLDER

```
/* Usage */
FOLDER *folder_fetch(
        char *folder_name,
        boolean fetch_attribute_list,
        boolean fetch_one2m_relation_list,
        boolean fetch_mto1_relation_list,
        boolean fetch_mto1_isa_relation_list );

/* Input */
char *application_name;
char *folder_name;
int insert_rows_number;
long row_count;
boolean insert_permission;
boolean update_permission;
boolean lookup_permission;
char *post_change_javascript;
boolean lookup_email_output;
char *session;
char *folder_form;
long row_access_count;
boolean lookup_before_drop_down;
char *notepad;
PROCESS *populate_drop_down_process;
PROCESS *post_change_process;
char *html_help_file_anchor;
LIST *non_primary_attribute_list;
boolean lookup_transmit_output;
boolean row_level_non_owner_forbid;
boolean row_level_non_owner_view_only;
boolean no_initial_capital;
char *schema_name;

/* Process */
LIST *attribute_list(
        folder_name );

LIST *folder_primary_attribute_name_list(
        attribute_list() );
LIST *primary_attribute_name_list(
        attribute_list() );

DICTIONARY *primary_data_dictionary;

LIST *folder_mto1_isa_related_folder_list(
        existing_related_folder_list,
        application_name,
        folder_name,
        role_name,
        recursive_level );

LIST *folder_append_isa_attribute_list(
        application_name,
        folder_name,
        folder_mto1_isa_related_folder_list,
        role_name );

LIST *folder_attribute_name_list(
        attribute_list );

LIST *folder_append_isa_attribute_name_list(
        folder_append_isa_attribute_list );
```



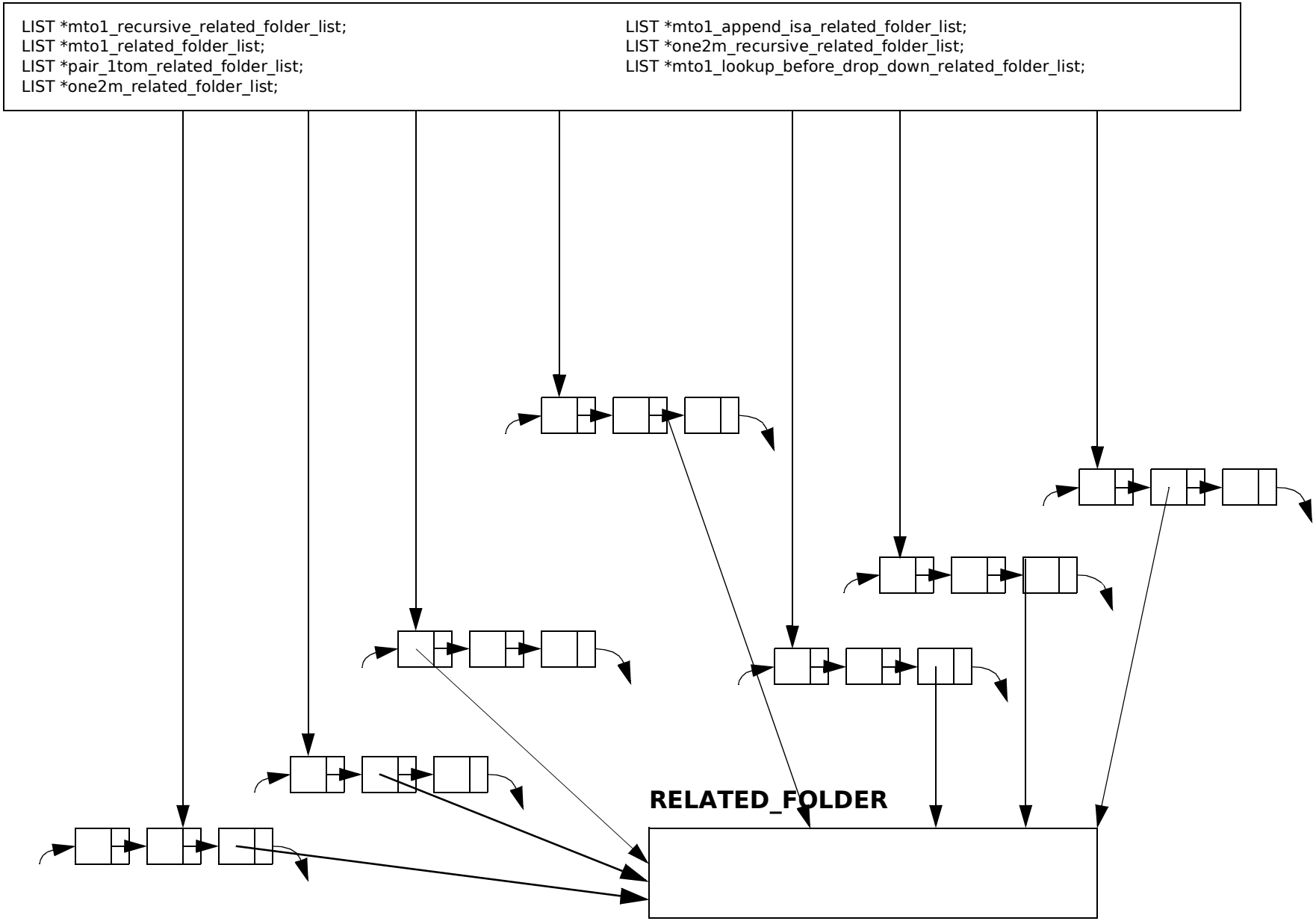
APPASERVER Data Structure

FOLDER (con’t.)



APPASERVER Data Structure

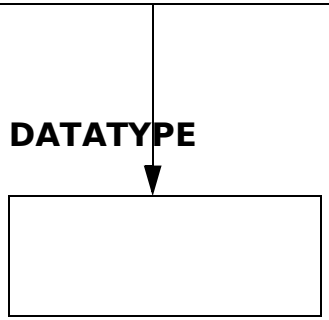
FOLDER (con’t.)



APPASERVER Data Structure

ATTRIBUTE

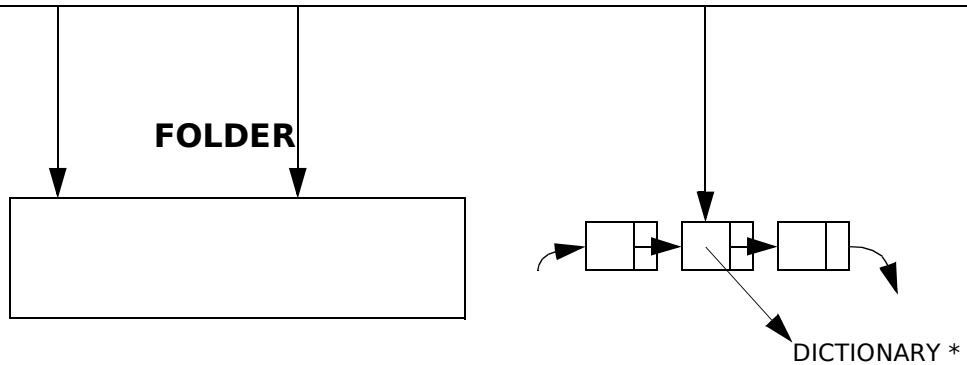
```
/* Input */
char *attribute_name;
char *datatype_name;
DATATYPE *datatype;
int width;
int float_decimal_places;
char *hint_message;
boolean lookup_required;
boolean insert_required;
char *post_change_javascript
char *on_focus_javascript_function
boolean additional_unique_index
boolean additional_index
boolean lookup_histogram_output
boolean lookup_time_chart_output
boolean appaserver;

/* Process */
char *folder_name;
int display_order;
boolean omit_insert_prompt;
boolean omit_insert;
int primary_key_index;
LIST *exclude_permission_list;

LIST *attribute_primary_name_list(
        attribute_list() );
```



APPASERVER Data Structure

RELATED_FOLDER

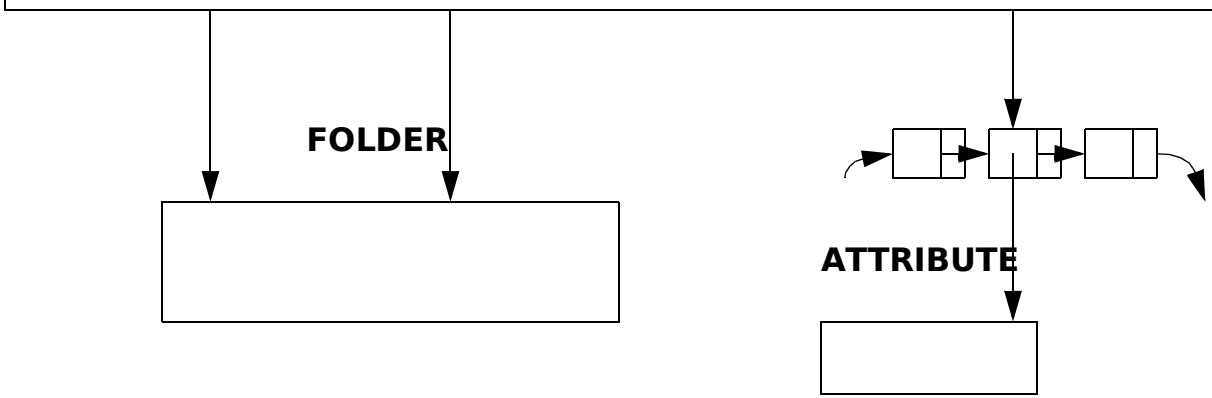
```
/* Input */
FOLDER *folder;
FOLDER *one2m_related_folder;
char *related_attribute_name;
boolean relation_type_isa;
int pair_1tom_order;
boolean null_1tom_upon_delete;
boolean ignore_output;
char *hint_message;
boolean copy_common_attributes;
boolean prompt_mto1_recursive;
boolean omit_1tom_detail;
char *join_where_clause;
boolean drop_down_multi_select;
boolean automatic_preselection;

/* Process */
LIST *foreign_attribute_name_list;
LIST *folder_foreign_attribute_name_list;
LIST *parent_foreign_attribute_name_list;
LIST *primary_data_list;
LIST *common_non_primary_attribute_name_list;
LIST *automatic_preselection_dictionary_list;
```



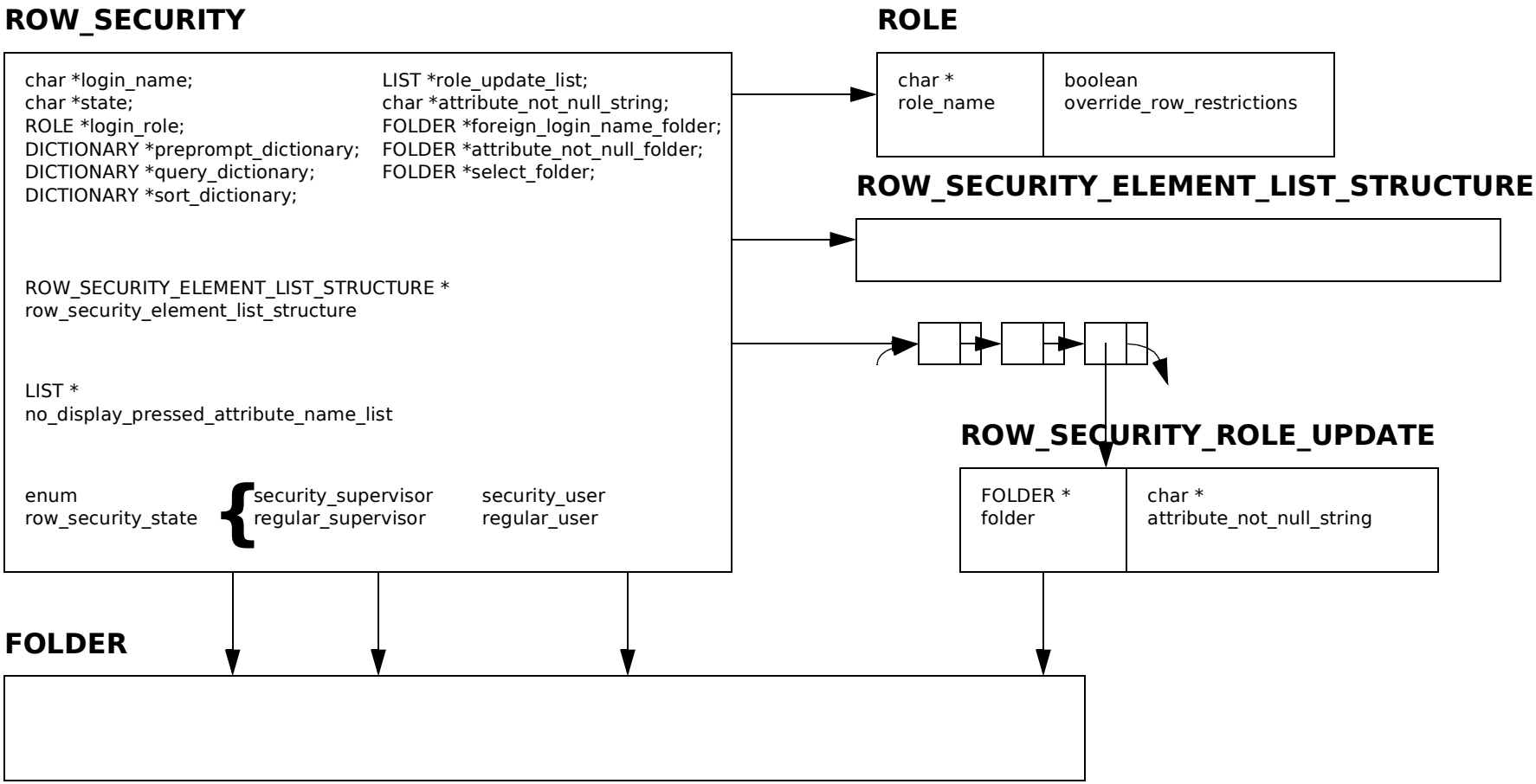
APPASERVER Data Structure

RELATION

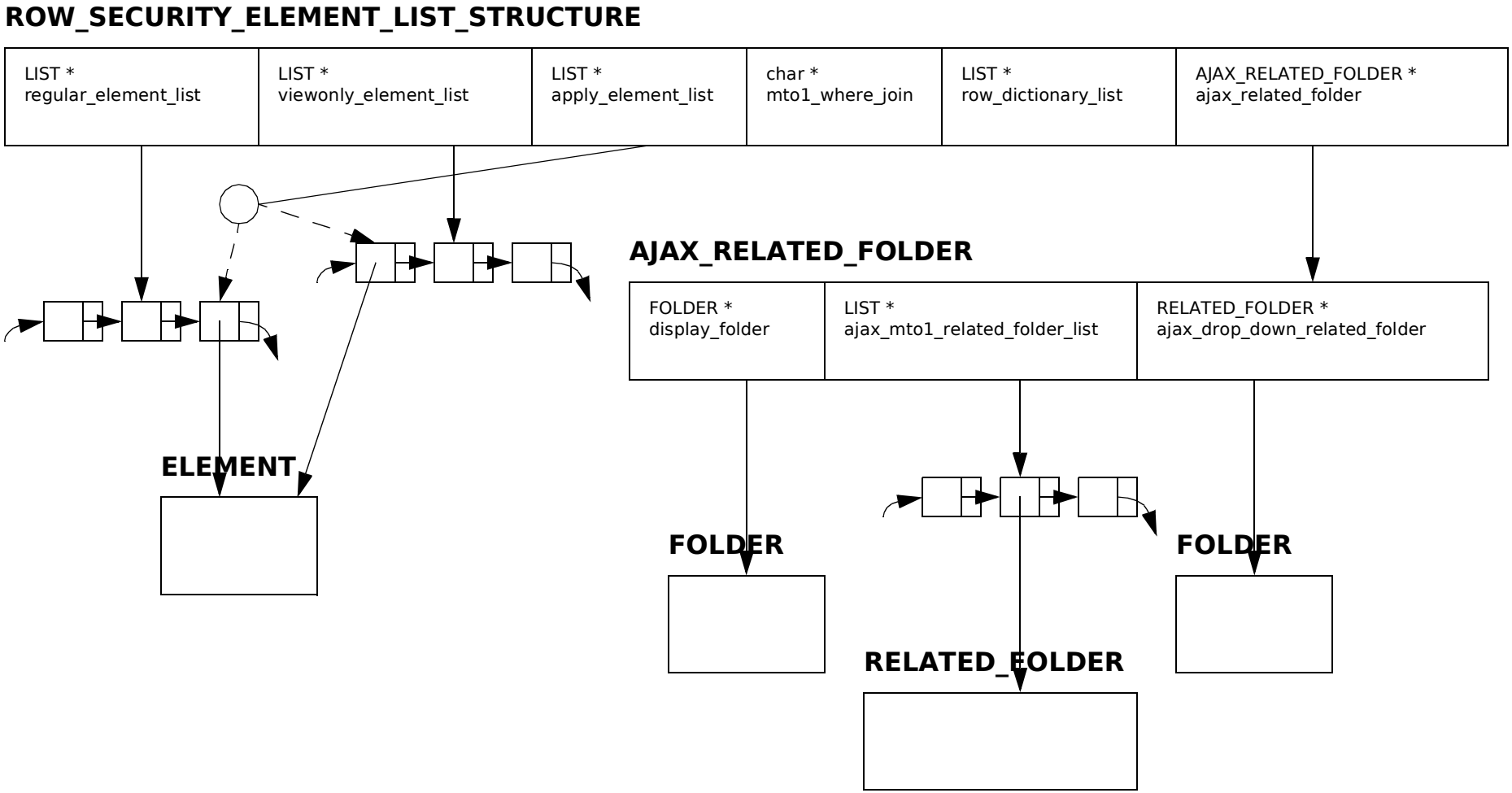
```
/* Input */
FOLDER *mto1_folder;
FOLDER *one2m_folder;
char *related_attribute_name;
int pair_1tom_order;
boolean omit_1tom_detail;
boolean prompt_mto1_recursive;
boolean relation_type_isa;
boolean copy_common_attributes;
boolean automatic_preselection;
boolean drop_down_multi_select;
boolean join_1tom_each_row;
boolean omit_lookup_before_drop_down;
boolean ajax_fill_drop_down;
char *hint_message;

/* Process */
LIST *relation_one2m_relation_list(
        char *one2m_folder_name );

LIST *relation_one2m_recursive_relation_list(
        LIST *relation_list /* in only */,
        char *one2m_folder_name );

boolean relation_is_primary_key_subset(
        LIST *foreign_attribute_name_list,
        LIST *mto1_primary_attribute_name_list );

LIST *relation_foreign_attribute_name_list(
        attribute_primary_name_list(
                one2m_folder->
                attribute_list )
                /* primary_foreign_attribute_name_list */,
        related_attribute_name,
        foreign_attribute_name_list(
                foreign_attribute_list_folder(
                        mto1_folder->folder_name
                        foreign_attribute_list() ) );

boolean ignore_output;

char *relation_join_where_clause();
```



APPASERVER Data Structure



APPASERVER Data Structure



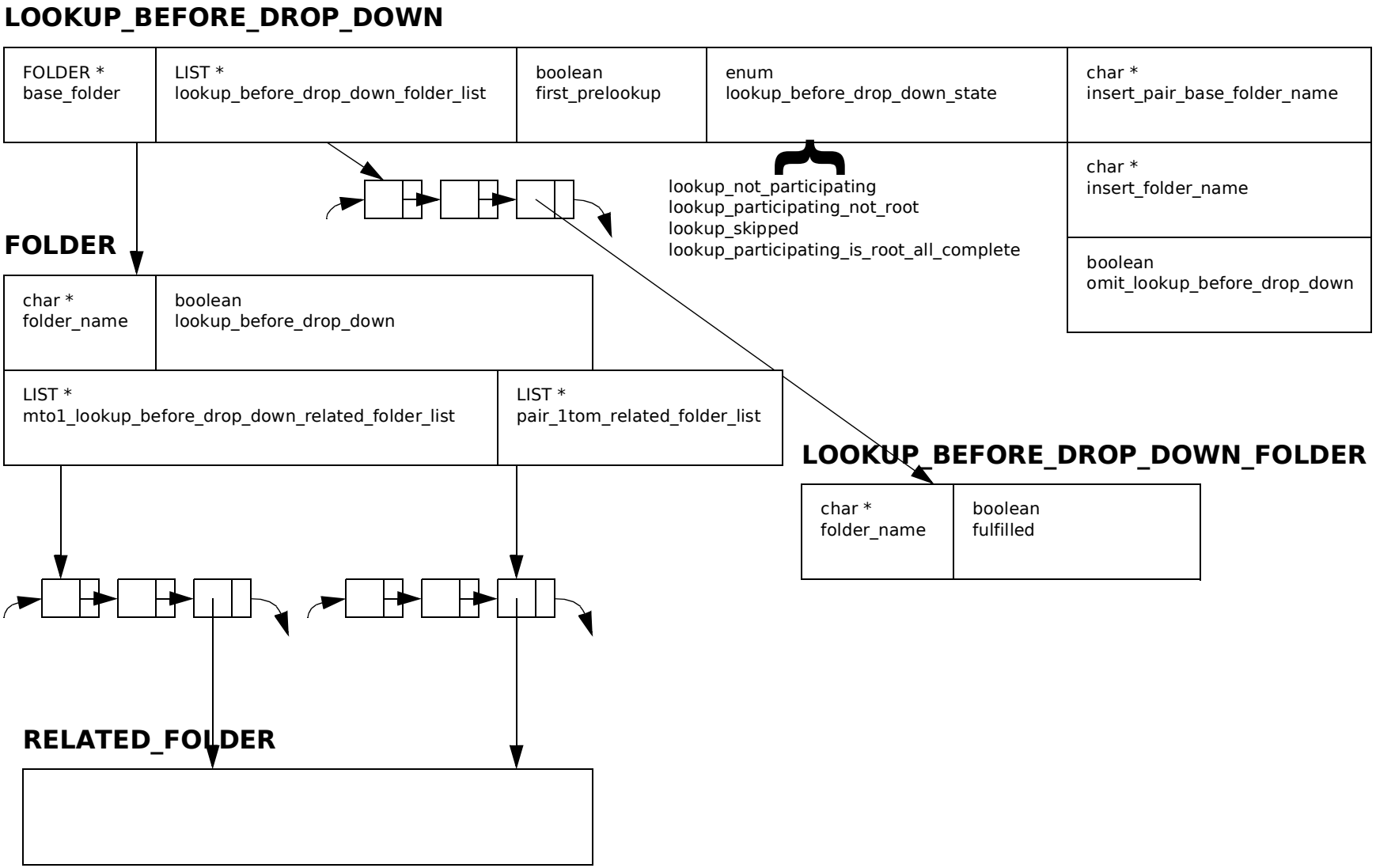
APPASERVER Data Structure

ELEMENT_APPASERVER

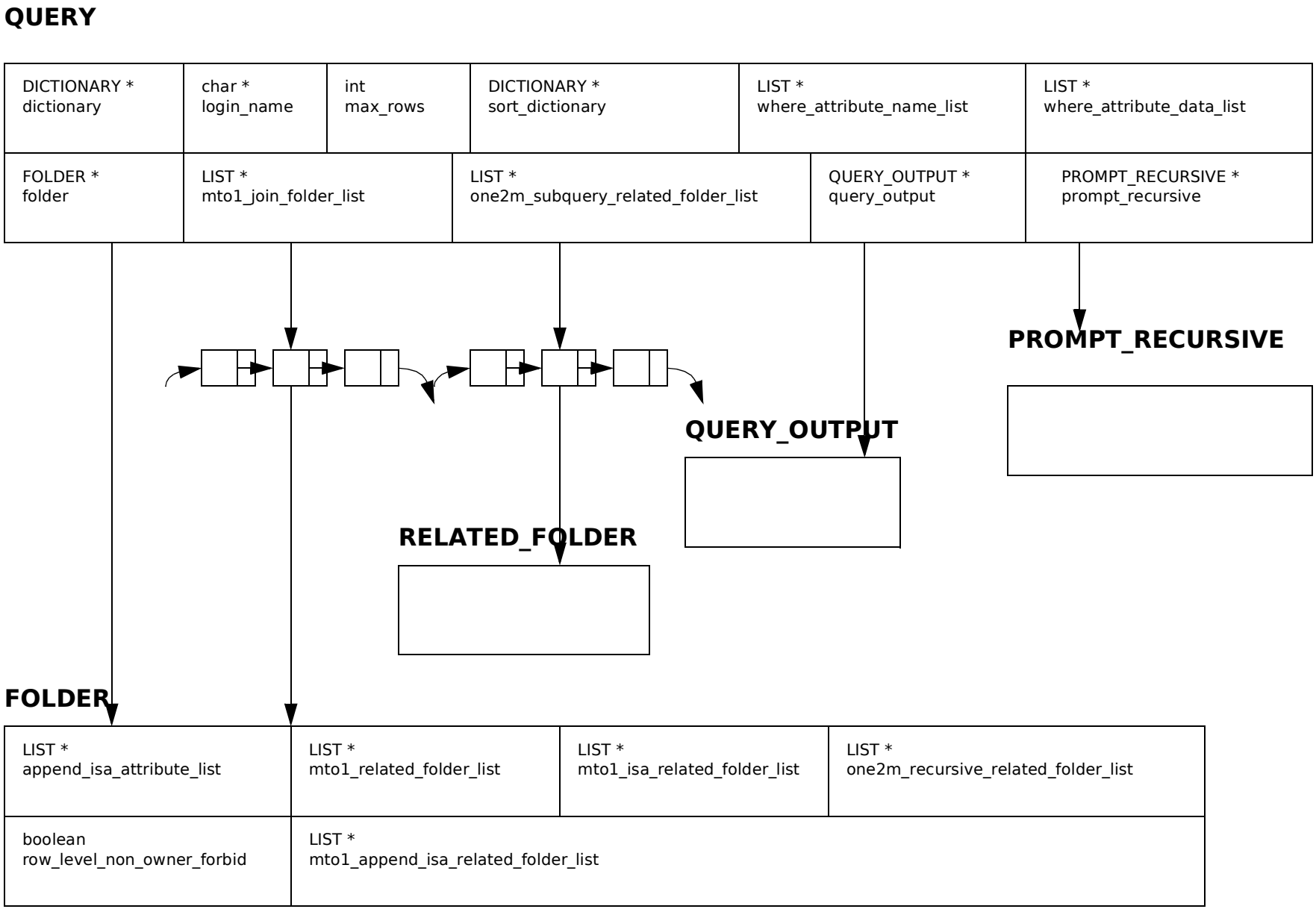
char * name	enum element_type element_type	boolean remember_keystrokes_ok	int tab_index
			ELEMENT_DROP_DOWN *drop_down ELEMENT_PUSH_BUTTON *push_button ELEMENT_RADIO_BUTTON *radio_button ELEMENT_NOTEPAD *notepad ELEMENT_LINEBREAK *linebreak ELEMENT_TEXT_ITEM *text_item ELEMENT_PASSWORD *password ELEMENT_PROMPT_DATA *prompt_data ELEMENT_HIDDEN *hidden ELEMENT_UPLOAD_FILENAME *upload_filename ELEMENT_HTTP_FILENAME *http_filename ELEMENT_ANCHOR *anchor ELEMENT_EMPTY_COLUMN *empty_column ELEMENT_REFERENCE_NUMBER *reference_number ELEMENT_NON_EDIT_TEXT *non_edit_text

drop_down
push_button
radio_button
notepad
password
non_edit_text
linebreak
text_item
upload_filename
prompt_data
prompt_data_plus_hidden
reference_number
hidden
empty_column
blank
element_date
element_current_date
element_time
element_current_time
http_filename
timestamp
table_opening
table_closing
prompt
anchor

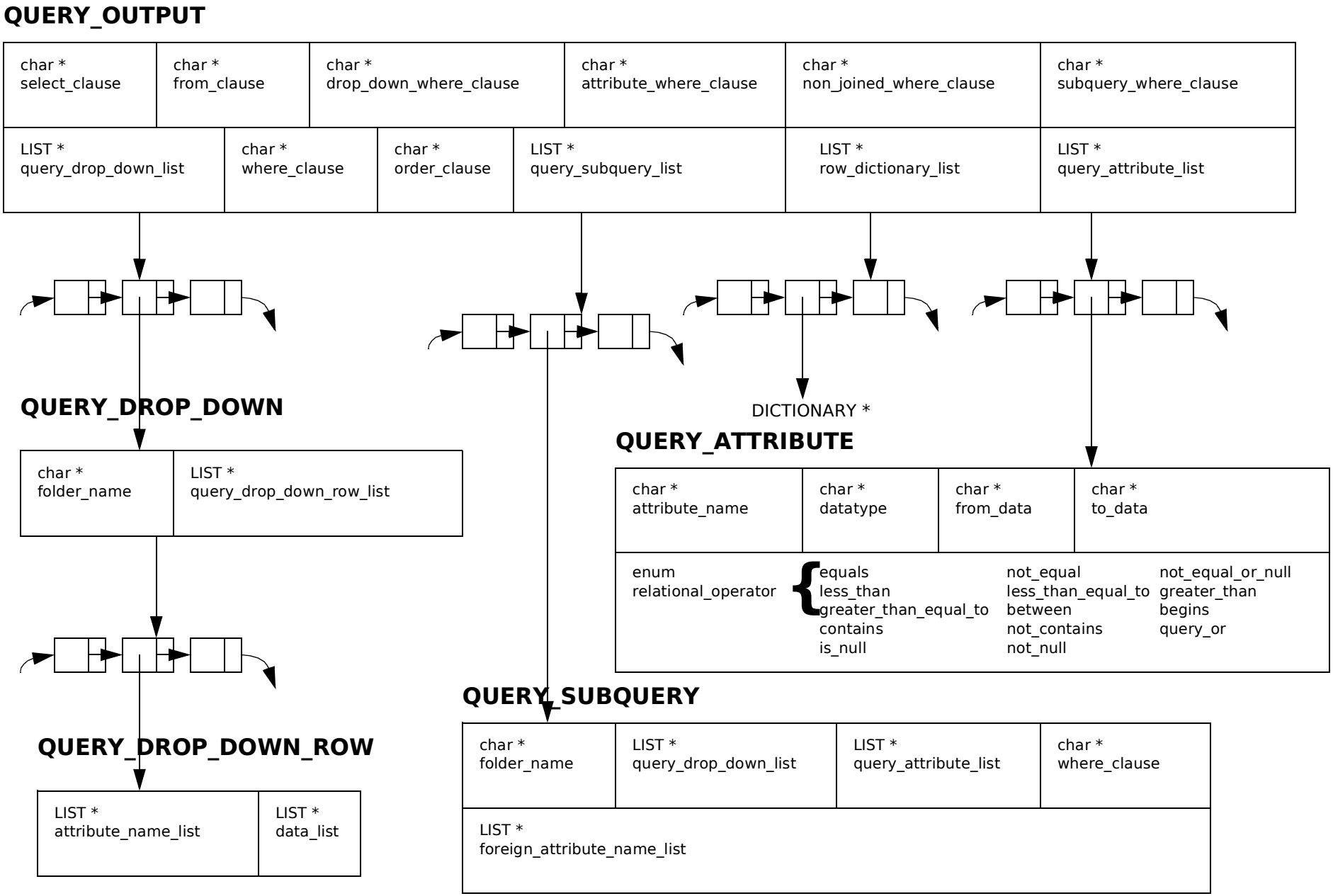
APPASERVER Data Structure



APPASERVER Data Structure



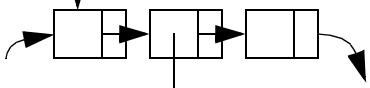
APPASERVER Data Structure



APPASERVER Data Structure

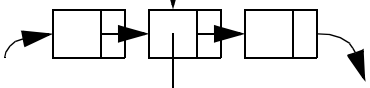
UPDATE_DATABASE

```
/* Input */
char *application_name;
char *session;
char *login_name;
char *role_name;
DICTIONARY *post_dictionary;
DICTIONARY *file_dictionary;

/* Process */
FOLDER *folder =
        folder_fetch(
                folder_name,
                1 /* fetch_attribute_list */,
                1 /* fetch_one2m_recursive_relation_list */,
                1 /* fetch_mto1_isa_recursive_relation_list */,
                0 /* not fetch_mto1_relation_list */,
                1 /* fetch_process */ );

LIST *update_database_update_row_list(
        /* Sets preupdate_$attribute_name for trigger */
        DICTIONARY *post_dictionary,
        DICTIONARY *file_dictionary,
        char *folder_name,
        folder->attribute_list,
        folder->one2m_recursive_relation_list,
        folder->post_change_process );
```



UPDATE_ROW

```
/* Usage */
UPDATE_ROW *update_database_update_row(
        /* Sets preupdate_$attribute_name for trigger */
        DICTIONARY *post_dictionary,
        DICTIONARY *file_dictionary,
        char *folder_name,
        LIST *attribute_list,
        LIST *one2m_recursive_relation_list,
        PROCESS *post_change_process,
        int row );

/* Process */
LIST *update_folder_list =
        update_database_update_folder_list(
                primary_update_folder->primary_data_list,
                primary_update_folder->changed_attribute_list,
                primary_update_folder->changed_primary_key,
                one2m_recursive_relation_list,
                mto1_isa_recursive_relation_list,
                primary_update_folder );

boolean changed_primary_key;
```



UPDATE_FOLDER

APPASERVER Data Structure

UPDATE_FOLDER

```
/* Usage */
LIST *update_database_update_folder_list(
        int row,
        LIST *primary_data_list,
        LIST *primary_changed_attribute_list,
        boolean changed_primary_key,
        LIST *one2m_recursive_relation_list,
        UPDATE_FOLDER *primary_update_folder );

UPDATE_FOLDER *update_primary_update_folder(
        /* Sets preupdate_$attribute_name for trigger */
        DICTIONARY *post_dictionary,
        DICTIONARY *file_dictionary,
        char *folder_name,
        LIST *attribute_list,
        PROCESS *post_change_process,
        int row );

UPDATE_FOLDER *update_secondary_update_folder(
        mto1_folder->folder_name,
        relation_foreign_attribute_name_list(),
        primary_data_list,
        primary_changed_attribute_list );

/* Process */
LIST *primary_attribute_name_list =
        attribute_primary_name_list();

LIST *relation_foreign_attribute_name_list =
        relation_foreign_attribute_name_list();

LIST *update_folder_changed_attribute_list(
        boolean *changed_primary_key,
        /* Sets preupdate_$attribute_name for trigger */
        DICTIONARY *post_dictionary,
        DICTIONARY *file_dictionary,
        LIST *attribute_list,
        int row );

LIST *update_folder_primary_data_list(
        DICTIONARY *file_dictionary,
        LIST *primary_attribute_name_list,
        int row );

LIST *update_folder_where_attribute_name_list(
        LIST *primary_attribute_name_list,
        LIST *relation_foreign_attribute_name_list );

LIST *update_folder_where_attribute_list(
        LIST *update_folder_where_attribute_name_list(),
        LIST *update_folder_primary_data_list() );

char *update_folder_where_clause(
        update_folder_where_attribute_list() );

int update_folder_count(
        update_folder_where_clause() );

boolean changed_primary_key;
```

WHERE_ATTRIBUTE

```
char *attribute_name;
char *data;
```

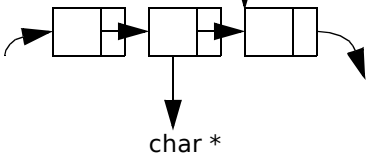
CHANGED_ATTRIBUTE

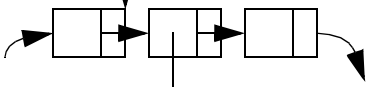
```
/* Usage */
LIST *update_changed_attribute_list(
        boolean *changed_primary_key,
        /* Sets preupdate_$attribute_name for trigger */
        DICTIONARY *post_dictionary,
        DICTIONARY *file_dictionary,
        LIST *attribute_list,
        int row );

CHANGED_ATTRIBUTE *update_changed_attribute(
        /* Sets preupdate_$attribute_name for trigger */
        DICTIONARY *post_dictionary,
        DICTIONARY *file_dictionary,
        ATTRIBUTE *attribute,
        int row );

CHANGED_ATTRIBUTE *update_changed_attribute_new(
        char *attribute_name,
        char *attribute_datatype,
        char *old_data,
        char *new_data );

char *changed_attribute_set_clause(
        char *attribute_name,
        char *attribute_datatype,
        char *new_data );

int changed_primary_key_index;
```

APPASERVER Data Structure

DELETE_DATABASE

char * application_name	boolean dont_delete_mto1_isa	char * sql_executable	char * login_name	char * role_name
FOLDER * folder	LIST * primary_attribute_data_list			LIST * delete_folder_list

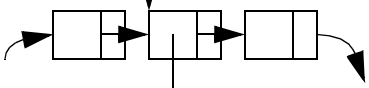


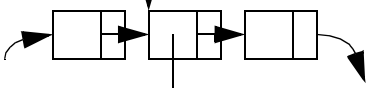


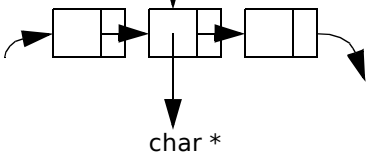
FOLDER

char * folder_name	boolean row_level_non_owner_forbid	boolean row_level_non_owner_view_only
LIST * one2m_recursive_related_folder_list	LIST * mto1_isa_related_folder_list	LIST * primary_attribute_name_list

DELETE_FOLDER





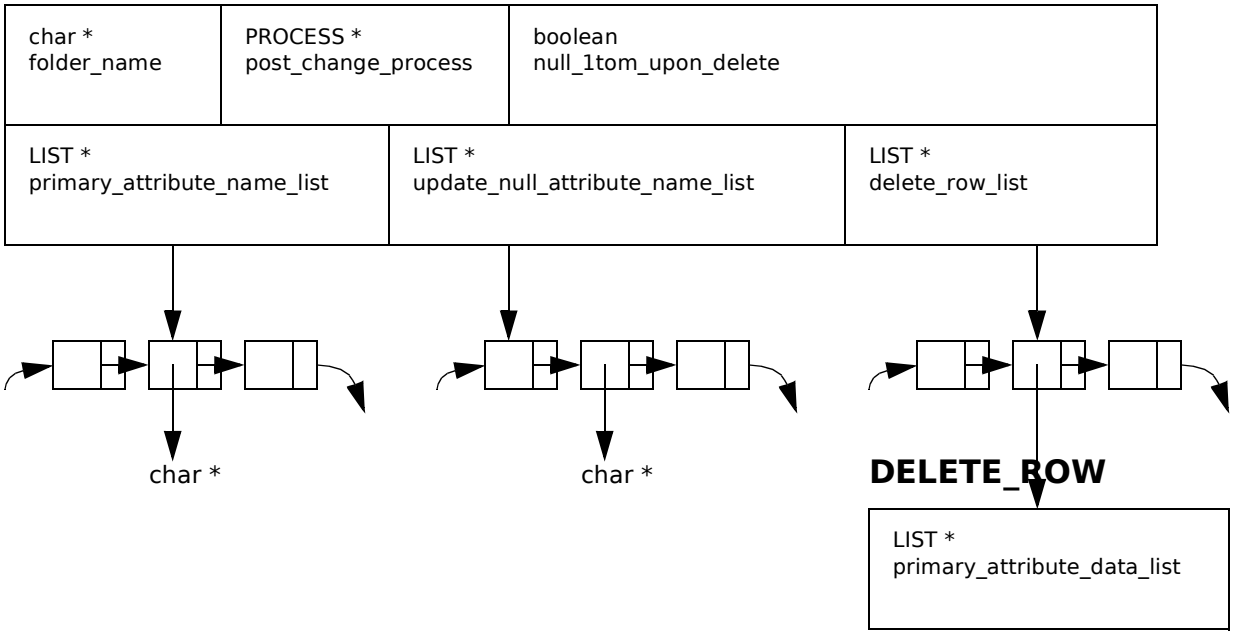


RELATED_FOLDER

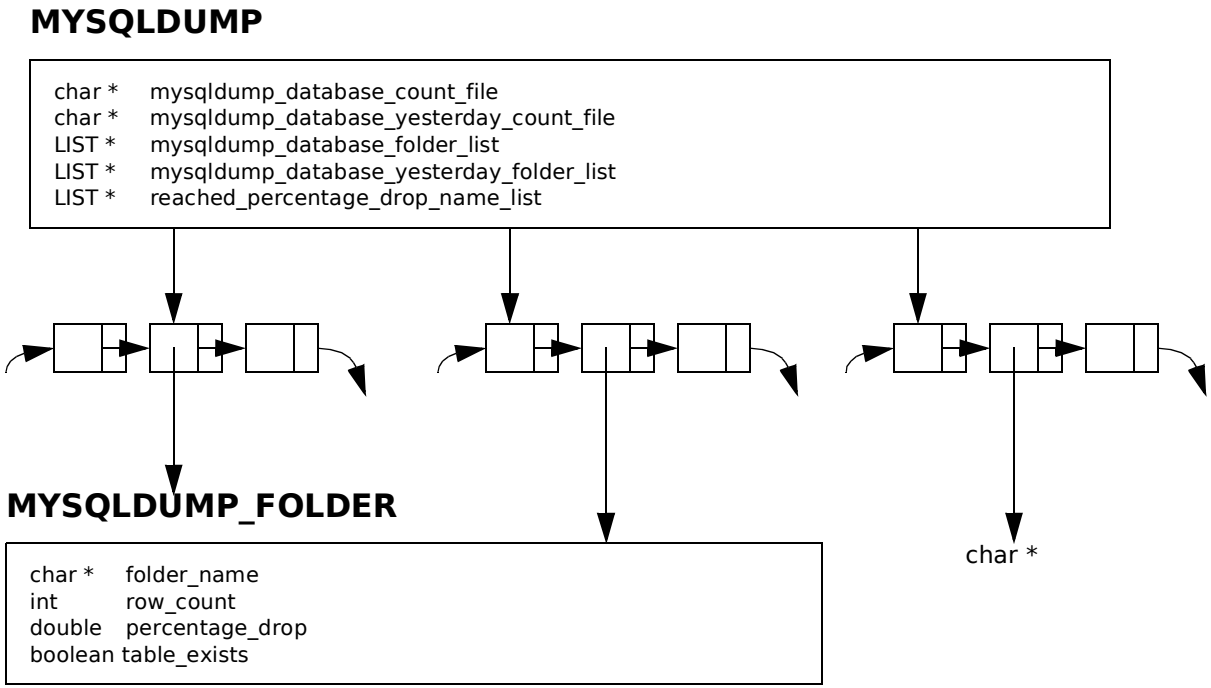
FOLDER * one2m_related_folder	FOLDER * one2m_related_folder	boolean null_1tom_upon_delete

APPASERVER Data Structure

DELETE_FOLDER



APPASERVER Data Structure



APPASERVER Data Structure

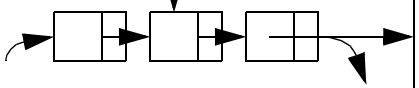
APPASERVER_USER

```
char *login_name
char *person_full_name
char *typed_in_password
char *database_password
char *user_date_format
boolean  frameset_menu_horizontal
LIST *role_list
LIST *session_list

enum password_function password_function; { no_encryption
                                            old_password_function
                                            password_function
                                            sha2_function

enum password_function
        appaserver_user_password_function(
                database_password )

char *appaserver_user_encryption_select(
                password_function
                typed_in_password )

char *appaserver_user_function_encrypted_password(
                application_name,
                typed_in_password,
                password_function )

boolean  appaserver_user_password_match(
                application_name,
                typed_in_password,
                database_password )

char *appaserver_user_mysql_version( void )

char *appaserver_user_version_encrypted_password(
                application_name,
                typed_in_password,
                mysql_version )

enum password_function
        appaserver_user_version_password_function(
                mysql_version )
```

APPASERVER Data Structure

PAIR_ONE2M

```
char *many_folder_name;
char *one_folder_name;

PAIR_ONE2M *pair_one2m_prompt_form_new(
        char *one_folder_name,
        char *keystrokes_save_function );

LIST *pair_one2m_prompt_form_folder_list(
        char *keystrokes_save_function,
        LIST *relation_one2m_pair_relation_list() );

char *pair_one2m_prompt_element_name(
        char *PAIR_ONE2M_PREFIX,
        char *folder_label );

boolean pair_one2m_participating(
        DICTIONARY *pair_one2m_dictionaryy );

PAIR_ONE2M *pair_one2m_post_new(
        DICTIONARY *pair_one2m_dictionary );

boolean pair_one2m_duplicate(
        char *MYSQL_DUPLICATE_ERROR_MESSAGE_KEY,
        char *message );

void pair_one2m_duplicate_set(
        DICTIONARY *pair_one2m_dictionary,
        char *PAIR_ONE2M_DUPLICATE_KEY );

boolean pair_one2m_inserted_duplicate(
        char *PAIR_ONE2M_DUPLICATE_KEY,
        DICTIONARY *pair_one2m_dictionary );

LIST *pair_one2m_fulfilled_folder_name_list(
        char *PAIR_ONE2M_FULFILLED_LIST_LABEL,
        DICTIONARY *pair_one2m_dictionary );

char *pair_one2m_folder_name(
        char *folder_label,
        DICTIONARY *pair_one2m_dictionary );

void pair_one2m_folder_set(
        DICTIONARY *pair_one2m_dictionary,
        char *folder_label,
        char *folder_name );

char *pair_one2m_next_folder_name(
        LIST *pair_one2m_fulfilled_folder_name_list(),
        LIST *relation_one2m_pair_relation_list(),
        char *one_folder_name,
        char *many_folder_name );

DICTIONARY  *pair_one2m_fulfilled_dictionary(
        DICTIONARY *pair_one2m_dictionary,
        char *PAIR_ONE2M_FULFILLED_LIST_LABEL,
        LIST *pair_one2m_fulfilled_folder_name_list );

/* Insert form */
PAIR_ONE2M *pair_one2m_insert_form_new(
        DICTIONARY *pair_one2m_dictionary );
```



PAIR_ONE2M_FOLDER

```
/* Usage */
pair_one2m_folder_new(
        char *many_folder_name );

char *many_folder_name;

char *pair_folder_element_copy_function(
        prompt_many_element_name,
        many_folder_name,
        pair_one2m_prompt_form_element_name(),
        folder_name );

char *pair_folder_button_string(
        many_folder_name,
        keystrokes_save_function,
        pair_folder_element_copy_function() );
```

APPASERVER Data Structure

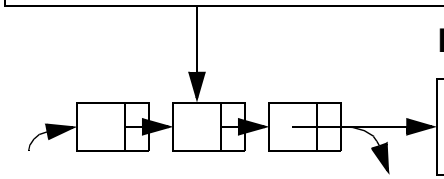
VERTICAL_NEW_BUTTON

```
char *many_folder_name;
char *one_folder_name;

VERTICAL_NEW_BUTTON *vertical_new_button_calloc(
        void );

char *vertical_new_button_one_element_name(
        char *VERTICAL_NEW_BUTTON_ONE_PREFIX,
        char *one_folder_name );

char *vertical_new_button_many_element_name(
        char *VERTICAL_NEW_BUTTON_MANY_LABEL );

ELEMENT_APPASERVER *vertical_new_button_element(
        char *one_folder_name,
        LIST *role_folder_insert_list,
        char *form_name,
        char *VERTICAL_NEW_BUTTON_ONE_PREFIX,
        char *VERTICAL_NEW_BUTTON_DISPLAY,
        char *onclick_keystrokes_save_string );

char *vertical_new_button_dictionary_one_folder_name(
        char *VERTICAL_NEW_BUTTON_ONE_PREFIX,
        DICTIONARY *non_prefixed_dictionary );

char *vertical_new_button_dictionary_folder_name(
        char *hidden_label,
        DICTIONARY *non_prefixed_dictionary );

void vertical_new_button_dictionary_set(
        DICTIONARY *non_prefixed_dictionary,
        char *hidden_label,
        char *folder_name );
```

APPASERVER Data Structure

INSERT_DATABASE

```
char *application_name;
char *session;
char *folder_name;
LIST *primary_attribute_name_list;
LIST *attribute_name_list;
LIST *data_list;
DICTIONARY *row_dictionary;
LIST *ignore_attribute_name_list;
boolean insert_row_zero_only;
boolean dont_remove_tmp_file;
int rows_inserted;
char *post_change_process;
char *login_name;
char *tmp_file_name;
FILE *tmp_file;
LIST *attribute_list;

LIST *attribute_data_list =
        insert_database_attribute_data_list(
                row_dictionary,
                row,
                ignore_attribute_name_list,
                primary_attribute_name_list,
                attribute_name_list,
                mto1_related_folder_list,
                common_non_primary_attribute_name_list,
                application_name,
                attribute_list );

LIST *insert_attribute_name_list =
        insert_database_attribute_name_list(
                insert_database_attribute_data_list() );
```



INSERT_DATABASE_ATTRIBUTE_DATA

```
char *attribute_name;
char *data;
```

APPASERVER Data Structure

PROCESS_GENERIC_OUTPUT

```
/* Input */
char *process_set_name;
char *process_name;
char *output_medium_string;
DICTIONARY *post_dictionary;

/* To retire */
boolean accumulate;

/* Usage */
PROCESS_GENERIC_OUTPUT *process_generic_output_calloc();

/* Process */
char *process_name =
        process_generic_output_process_name(
                this->process_set_name,
                post_dictionary );

char *value_folder_name =
        process_generic_output_value_folder_name(
                this->process_name,
                this->process_set_name );

PROCESS_GENERIC_VALUE_FOLDER *value_folder =
        process_generic_value_folder_fetch(
                this->value_folder_name );

PROCESS_GENERIC_PARAMETER *parameter =
        process_generic_parameter_parse();

char *process_generic_output_system_string(
        process_generic_datatype_measurement_select(
                this->value_folder->foreign_folder->foreign_attribute_name_list,
                this->value_folder->date_attribute,
                this->value_folder->time_attribute,
                this->value_folder->->value_attribute,
                this->value_folder_name,
                process_generic_datatype_measurement_where(),
                this->parameter->aggregate_level,
                this->parameter->aggregate_statistic,
                this->
                        value_folder->
                        foreign_folder->
                        foreign_attribute_name_list_length,
                this->parameter->end_date_string,
                this->parameter->accumulate );
```

PROCESS_GENERIC_PARAMETER

```
/* Attributes */
char *output_medium_string;
enum process_generic_output_medium output_medium;
char delimiter;
char *begin_date;
char *end_date;
enum aggregate_level aggregate_level;
enum aggregate_statistic aggregate_statistic;
boolean accumulate;
char *email_address;
char *units_converted;
char *end_date_suffix;
pidt process_id;

/* Usage */
PROCESS_GENERIC_PARAMETER *process_generic_parameter_parse(
        char *output_medium_string,
        post_dictionary );
```

{ text_file, output_medium_stdout, table, spreadsheet

PROCESS_GENERIC_VALUE_FOLDER

APPASERVER Data Structure

PROCESS_GENERIC_VALUE_FOLDER

```
/* Attributes */
char *value_folder_name;
char *date_attribute_name;
char *time_attribute_name;
char *value_attribute_name;
char *datatype_folder_name;
char *foreign_folder_name;
LIST *primary_attribute_name_list;
PROCESS_GENERIC_FOREIGN_FOLDER *foreign_folder;

/* Usage */
PROCESS_GENERIC_VALUE_FOLDER *value_folder =
        process_generic_value_folder_fetch(
                process_generic_output->value_folder_name );

/* Process */
PROCESS_GENERIC_DATATYPE_FOLDER *datatype_folder =
        process_generic_datatype_folder_fetch(
                this->datatype_folder_name );

PROCESS_GENERIC_DATATYPE_FOREIGN_FOLDER *foreign_folder =
        process_generic_foreign_folder_fetch(
                this->foreign_folder_name,
                process_generic_output->post_dictionary );
```

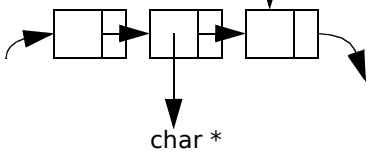
PROCESS_GENERIC_FOREIGN_FOLDER

```
char *foreign_folder_name;
FOLDER *foreign_folder;

LIST *foreign_attribute_name_list =
        foreign_folder->primary_attribute_name_list;

LIST *foreign_attribute_data_list =
        dictionary_list_fetch(
                process_generic_output->post_dictionary,
                this->foreign_attribute_name_list );

int foreign_attribute_name_list_length =
        list_length(
                this->foreign_attribute_name_list );
```

PROCESS_GENERIC_DATATYPE_FOLDER

FOLDER

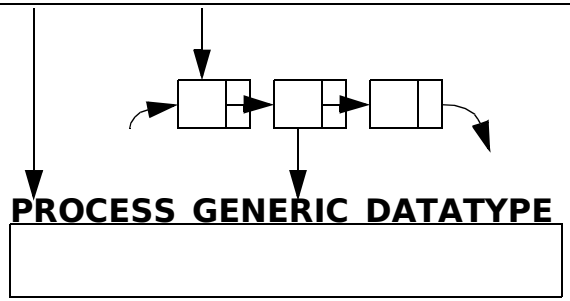
```
char *folder_name;
LIST *primary_attribute_name_list
```



APPASERVER Data Structure

PROCESS_GENERIC_DATATYPE_FOLDER

```
/* Attributes */
char *datatype_folder_name;
boolean exists_aggregation_sum;
boolean exists_bar_graph;
boolean exists_scale_graph_zero;
boolean exists_unit;

/* Usage */
PROCESS_GENERIC_DATATYPE_FOLDER *
        /* Also fetches this->datatype */
        process_generic_datatype_folder_fetch(
                process_generic_output->value_folder->datatype_folder_name,
                process_generic_output->post_dictionary );

LIST *primary_attribute_name_list =
        process_generic_datatype_folder_primary_attribute_name_list(
                datatype_folder_name );

LIST *primary_attribute_data_list =
        process_generic_datatype_folder_primary_data_list(
                this->primary_attribute_name_list,
                process_generic_output->post_dictionary );

char *select =
        process_generic_datatype_folder_select(
                boolean exists_aggregation_sum,
                boolean exists_bar_graph,
                boolean exists_scale_graph_zero,
                boolean exists_unit );

char *where =
        process_generic_datatype_folder_where(
                this->primary_attribute_name_list,
                this->primary_attribute_data_list );

PROCESS_GENERIC_DATATYPE *datatype =
        process_generic_datatype_fetch(
                this->datatype_folder_name,
                this->select,
                this->where(),
                this->exists_aggregation_sum,
                this->exists_bar_graph,
                this->exists_scale_graph_zero,
                this->exists_unit );

LIST *process_generic_datatype_compare_list(
        process_generic_output->post_dictionary,
        this->exists_aggregation_sum,
        this->exists_bar_graph,
        this->exists_scale_graph_zero,
        this->exists_unit );
```

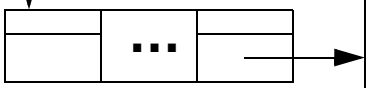


APPASERVER Data Structure

PROCESS_GENERIC_DATATYPE

```
/* Attributes */
char *datatype_name;
boolean aggregation_sum;
boolean bar_graph;
boolean scale_graph_zero;
char *unit;

/* Usage */
PROCESS_GENERIC_DATATYPE *process_generic_datatype_fetch(
        char *process_generic_datatype_folder->datatype_folder_name,
        char *process_generic_datatype_folder_select(),
        char *process_generic_datatype_folder_where(),
        boolean process_generic_datatype_folder->exists_aggregation_sum,
        boolean process_generic_datatype_folder->exists_bar_graph,
        boolean process_generic_datatype_folder->exists_scale_graph_zero,
        boolean process_generic_datatype_folder->exists_unit );

/* Process */

PROCESS_GENERIC_DATATYPE *process_generic_datatype_parse(
        char *system_string,
        boolean exists_aggregation_sum,
        boolean exists_bar_graph,
        boolean exists_scale_graph_zero,
        boolean exists_unit );

char *process_generic_datatype_date_where(
        char *process_generic_value_folder_fetch()->date_attribute,
        char *begin_date_string,
        char *end_date_string );

char *process_generic_datatype_measurement_where(
        LIST *foreign_attribute_name_list,
        LIST *foreign_attribute_data_list,
        char *process_generic_datatype_date_where() );

char *process_generic_datatype_measurement_select(
        LIST *foreign_attribute_name_list,
        char *date_attribute,
        char *time_attribute,
        char *value_attribute );

char *process_generic_datatype_measurement_heading(
        process_generic_datatype_measurement_select(),
        accumulate_boolean );
```

key = date or
key = date^time



PROCESS_GENERIC_VALUE

```
char *date;
char *time;
char *unit;
double value;
boolean is_null;
double accumulate

PROCESS_GENERIC_VALUE *process_generic_value_parse(
        double *accumulate_value,
        char *input,
        char *time_attribute,
        boolean unit_value_folder,
        boolean accumulate_boolean,
        enum aggregate_level );
```